

Spatial-JEPA: Geometry-Aware Self-Supervised Learning from RGBD via Cross-Attention Fusion

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Abstract—Many robotic systems, including manipulation, navigation, surgical scopes, and AR/VR, operate with calibrated RGB-D sensors at inference time, yet self-supervised visual encoders remain dominated by RGB-only objectives that learn appearance rather than metric scene geometry. We present Spatial-JEPA, a Joint Embedding Predictive Architecture designed as a geometry-grounded RGB-D encoder for these systems: a dual-stream architecture in which depth tokens query RGB tokens through asymmetric cross-attention to produce fused patch representations whose pairwise distances correlate with real-world 3D Euclidean distances. Against three RGB-D fusion baselines trained under identical recipes (concat fusion, symmetric cross-attention, and depth-aware masking), the asymmetric design is the decisive choice: it lifts per-image Spearman ρ from 0.256/0.273 to 0.571 on NYUv2 without any geometric supervision. Downstream, a frozen-encoder depth probe reaches AbsRel=0.046 at 224×224 and AbsRel=0.037 at full 640×480 resolution ($11 \times$ a matched-regime RGB-only baseline at 0.421), and zero-shot transfer to two medical RGB-D domains (synthetic and real-phantom colonoscopy) shows that the geometric prior generalises to deployment domains unrelated to indoor scene pretraining. Spatial-JEPA pretrained on HyperSim (4,484 synthetic frames) achieves $\rho = 0.611$ on NYUv2 zero-shot, surpassing the NYUv2-trained model, while the matched RGB baseline gains nothing. For RGB-only deployment, two routes recover the geometric prior: chaining the trained encoder with a frozen monocular depth predictor matches the GT-depth result ($\rho = 0.578$), and cross-modal distillation into a ~ 25 M-parameter RGB-only student recovers 79% of the per-image prior.

Index Terms—self-supervised learning, RGBD, geometric representations, depth estimation, cross-attention fusion

I. INTRODUCTION

Autonomous systems operating in physical environments, surgical robots, warehouse automation, assistive navigation, must reason about *where* objects are in metric 3D space, not merely *what* they look like. Dominant self-supervised vision models, including I-JEPA [2], DINO [7], and MAE [8] still learn representations organised by visual appearance: two patches of bare wall at 1 m and 5 m depth produce similar embeddings, while two patches of different materials at the same depth produce dissimilar ones. This is appropriate for 2D recognition but is a fundamental liability for spatial reasoning.

The Joint Embedding Predictive Architecture (JEPA) paradigm [1], instantiated for images in I-JEPA and video in V-JEPA [3], learns by predicting abstract representations of masked regions from visible context, avoiding pixel reconstruction. Recent extensions bring JEPA to 3D: Point-JEPA [5] applies the objective to object-level point clouds, and Locate 3D [6] extends it to sensor point clouds pre-featurised

with 2D foundation models. However, neither explicitly learns metric scene geometry, Point-JEPA operates at object scale without metric analysis, and Locate 3D inherits 2D model biases by preprocessing with CLIP [20] and DINO [7] rather than learning from raw geometry.

We present **Spatial-JEPA**, an RGB-D self-supervised encoder for robotic systems that have calibrated depth at inference time. Spatial-JEPA extends JEPA to scene-level RGB-D data via a cross-attention fusion module: depth tokens (providing metric queries) attend to RGB tokens (providing appearance context), producing fused patch representations that are both appearance-aware and geometrically grounded. The contribution is the fusion mechanism, not a depth-free RGB representation: depth is consumed at every forward pass, as it is in the target deployment settings (mobile manipulators, surgical endoscopes, AR/VR, RGB-D SLAM front-ends). The key insight is architectural, not merely adding depth as a fourth input channel, but preserving the asymmetry between metric depth and photometric colour through separate encoding streams and asymmetric cross-attention. We verify this against three RGB-D fusion baselines trained under identical recipes.

Contributions:

- 1) **Architecture.** A dual-stream RGB-D encoder with asymmetric cross-attention fusion, verified against three RGB-D fusion baselines (concat, symmetric cross-attention, and depth-aware masking only) trained under identical recipes. To our knowledge this is the first JEPA variant designed to learn explicit metric scene geometry from raw RGB-D data.
- 2) **Downstream evaluation.** A frozen-encoder depth probe (AbsRel=0.046 at 224×224 , AbsRel=0.037 at full 640×480 resolution with interpolated positional embeddings, $11 \times$ a matched-regime RGB-only baseline) and zero-shot transfer to two medical RGB-D domains (SimCol3D, C3VD).
- 3) **Latent-geometry analysis.** Per-image $\rho = 0.571$ on NYUv2 with cross-attention vs. 0.273 with concat fusion: the asymmetric design is the source of the gain, not depth as input.
- 4) **Mechanistic analysis + depth-free deployment.** Three controlled experiments isolate where the geometry lives: in the embedding space it is depth-conditional, not baked into the RGB weights. Two deployment routes recover the prior on RGB-only hardware: chaining a frozen monocular depth predictor (matches the GT-depth result,

$\rho = 0.578$) and cross-modal distillation into an RGB-only student ($\rho = 0.454$, 79% of the prior).

- 5) **Depth dropout.** A training strategy that improves cross-image ρ $3\times$ with negligible cost to per-image geometry, useful for partial-depth pipelines.
- 6) **Cross-dataset generalisation.** Spatial-JEPA trained on HyperSim achieves $\rho = 0.611$ on NYUv2 zero-shot, surpassing the NYUv2-trained model (0.571), while RGB-only baselines gain nothing.

II. RELATED WORK

JEPA Family. I-JEPA [2] learns image representations via masked patch prediction in latent space. V-JEPA [3] extends this to video via spatio-temporal masking. LeJEPA [4] establishes that the theoretically optimal embedding distribution for JEPA-family models is an isotropic Gaussian $\mathcal{N}(0, \mathbf{I})$, providing the SIGReg regulariser we adopt for domain adaptation. Point-JEPA [5] applies JEPA to object-level point clouds but does not analyse latent geometry or study metric downstream tasks. Spatial-JEPA addresses what all prior JEPA work leaves open: scene-scale metric geometry learning from raw RGBD with mechanistic analysis.

3D self-supervised learning. Point-BERT [9] and Point-MAE [10] apply masked reconstruction to point clouds, generative objectives that do not produce the predictive latent representations studied here. DepthContrast [11] applies contrastive learning to LiDAR point clouds. These methods operate at object scale and do not study metric geometry.

Multimodal fusion. Methods that fuse RGB and depth via concatenation or symmetric attention [17], [18] exist in supervised settings but have not been applied to predictive SSL. We study asymmetric fusion, depth queries RGB, and show it is the decisive mechanism over both concatenation and depth-aware masking.

III. METHOD

A. Architecture

Given an RGB image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ and aligned depth map $\mathbf{D} \in \mathbb{R}^{H \times W}$, we seek patch representations $\mathbf{z} \in \mathbb{R}^{N \times d}$ whose pairwise distances correlate with pairwise 3D Euclidean distances of patch centroids back-projected from $\mathbf{D}$. No explicit metric supervision is provided.

RGB Stream. A ViT-Small [12] (depth=6, heads=6, $d=384$, patch_size=16) processes the 224×224 image into $N = 196$ patch tokens $\mathbf{r} \in \mathbb{R}^{196 \times 384}$. Each 16×16 patch is projected by a linear embedding layer; learnable positional embeddings are added. ($\sim 21\text{M}$ parameters.)

Depth Stream. A shallower ViT (depth=3, heads=6, $d=384$, in_channels=1) processes the normalised depth map: $\mathbf{d} \in \mathbb{R}^{196 \times 384}$. Positional embeddings are shared with the RGB stream. The reduced depth reflects the geometric simplicity of depth relative to appearance. ($\sim 3.2\text{M}$ parameters.)

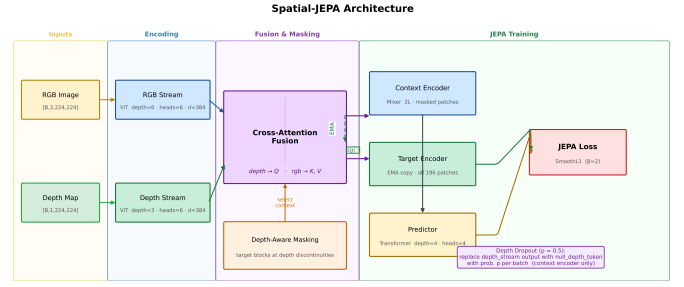


Fig. 1. Spatial-JEPA architecture. RGB and depth streams process the same 224×224 spatial grid into 196 patch tokens each. Depth tokens query RGB tokens via asymmetric cross-attention to produce fused representations. The context encoder processes unmasked fused tokens; the predictor recovers masked target embeddings. The target encoder (EMA copy) always sees all 196 patches and provides stable geometric training targets.

Cross-Attention Fusion. Depth tokens attend to RGB tokens via multi-head cross-attention (pre-norm, residual on depth tokens):

$$\mathbf{f} = \mathbf{d} + \text{Attn}(\text{LN}(\mathbf{d}), \text{LN}(\mathbf{r}), \text{LN}(\mathbf{r})) \quad (1)$$

$$\mathbf{f} = \mathbf{f} + \text{MLP}(\text{LN}(\mathbf{f})) \quad (2)$$

Depth provides metric queries (Q); RGB provides appearance keys and values (K, V). Each fused token carries the spatial position encoded by depth, enriched with visual appearance from RGB. This differs from symmetric fusion [17], [18], where neither modality is privileged, and from early fusion (depth as a fourth channel), where spatial encoding is entangled from the first layer. ($\sim 0.6\text{M}$ parameters.)

Context and Target Encoders. The context encoder processes fused tokens at *unmasked* positions through a shallow mixing transformer (depth=2). The target encoder is an EMA copy of the full dual-stream pipeline, updated as:

$$\theta_{\text{tgt}} \leftarrow m \theta_{\text{tgt}} + (1 - m) \theta_{\text{ctx}} \quad (3)$$

with momentum m cosine-annealed from 0.996 to 1.0 over training. The target encoder processes all 196 patches, providing stable geometric targets uncorrupted by masking.

Predictor. A narrow transformer (depth=4, heads=4, $d=384$, FFN hidden=192) takes context embeddings at unmasked positions plus learnable positional tokens at masked positions, and predicts target embeddings for masked patches. ($\sim 1.5\text{M}$ parameters.)

Parameter totals. Trainable: $\sim 23.8\text{M}$ (context path); total including frozen target EMA: $\sim 45.7\text{M}$. The full architecture is illustrated in Fig. 1.

B. Depth-Aware Masking

Target blocks are selected to preferentially sample patches at **depth-discontinuity boundaries**. We compute a per-patch discontinuity score as the Laplacian response of $\mathbf{D}$ averaged within each 16×16 patch cell. Four target masking blocks are biased toward high-discontinuity regions, each covering 15–20% of the 196-patch grid (29–39 patches per block); with partial overlaps the union yields ≈ 95 masked patches. Context patches (≈ 101) form the complement. An ablation confirms

depth-aware masking contributes to training stability but not independently to metric structure (Sec. IV-B).

C. Training Objective

The JEPA prediction loss is SmoothL1 between predictor outputs and stop-gradient target encoder outputs for each masked patch:

$$\mathcal{L}_{\text{JEPA}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \text{SmoothL1}(\hat{\mathbf{z}}_i, \text{sg}(\mathbf{z}_i^{\text{tgt}})) \quad (4)$$

where $\mathcal{M}$ is the set of masked patch indices and $\text{sg}(\cdot)$ is stop-gradient. SmoothL1 (Huber loss, $\delta = 1$) is more robust than MSE to outlier predictions during early training.

For domain adaptation, we augment with SIGReg [4]:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{JEPA}} + \lambda \mathcal{L}_{\text{SIGReg}} \quad (5)$$

SIGReg promotes an isotropic Gaussian $\mathcal{N}(0, \mathbf{I})$ embedding distribution by matching the empirical characteristic function of patch embeddings to that of a standard Gaussian via $r = 128$ random 1D projections. The isotropic Gaussian minimises the worst-case downstream prediction risk for JEPA-family models [4]. During standard pretraining $\lambda = 0$; for domain-adaptation runs $\lambda = 0.1$.

D. Depth Dropout

During pretraining, we optionally replace the depth stream output with a learnable **null depth token** $\mathbf{n} \in \mathbb{R}^{196 \times 384}$ with probability p per batch. The context encoder sees real depth $(1-p)$ of the time and the null token p of the time. The target encoder always uses real depth, providing stable geometric targets. This forces the RGB pathway to encode scene-level structure on null-depth batches, while both pathways contribute on full-depth batches. $\mathbf{n}$ is updated jointly with all other parameters.

IV. EXPERIMENTS

Setup. All models train on NYU Depth v2 [13] (795 train / 654 val, 224×224). Depth maps are normalised to $[0, 1]$ via uint16-to-float conversion; at evaluation, multiplied by 10.0 to recover metres. Camera intrinsics for the 224×224 crop: $f_x = f_y = 518.86$, $c_x = 117.58$, $c_y = 125.74$. AdamW ($\beta = 0.9/0.95$, $w_d = 0.05$), cosine LR with 10-epoch linear warmup, gradient clipping at 1.0, 100 epochs, batch size 16. All training on Apple M4 (MPS, PyTorch 2.9.1). **All baselines train from scratch on identical data with no ImageNet or external pretraining.** This equal-data, equal-compute design isolates the contribution of the training objective and fusion mechanism independent of pretraining scale.

Baseline coverage. Table I summarises every method evaluated in this paper, grouped by inference-time input modality. Within the RGB-D group, three fusion designs are trained from scratch under the identical recipe described above: concatenation (Ablation C), symmetric cross-attention (Ablation D), and asymmetric cross-attention (Spatial-JEPA). This makes the comparison apples-to-apples for the fusion-mechanism question.

TABLE I

METHODS EVALUATED IN THIS PAPER, GROUPED BY INFERENCE-TIME INPUT. ALL RGB-D VARIANTS TRAIN UNDER AN IDENTICAL SSL RECIPE; THE ONLY DIFFERENCE IS THE FUSION MECHANISM, WHICH IS WHAT WE ABLATE.

Method	Input @ inf.	Pretraining data	Role
CLIP [20]	RGB	WIT-400M	Internet baseline
DINOv2 [21]	RGB	LVD-142M	Internet baseline
Supervised RGB (depth)	RGB	NYUv2 + GT depth	Supervised reference
I-JEPA [2]	RGB	NYUv2	RGB-SSL baseline
Ablation B (depth-aware mask)	RGB-D	NYUv2	RGBD-SSL ablation
Ablation C (concat fusion)	RGB-D	NYUv2	RGBD-SSL ablation
Ablation D (symmetric x-attn)	RGB-D	NYUv2	RGBD-SSL baseline
Spatial-JEPA (asym. x-attn)	RGB-D	NYUv2	Ours

TABLE II

LATENT GEOMETRY, NYUv2 VAL (654 IMAGES).

Model	Training data	Per-image ρ	Cross-image ρ
CLIP ViT-B/16 [20]	WIT-400M (internet)	0.195 ± 0.087	0.040
DINOv2 ViT-S/14 [21]	LVD-142M (internet)	N/A [†]	0.093
I-JEPA (RGB SSL, 100ep)	NYUv2 (795 images)	0.256 ± 0.168	0.025
Spatial-JEPA (RGBD SSL, 100ep)	NYUv2 (795 images)	0.571 ± 0.166	0.104

[†]DINOv2 ViT-S/14 uses patch_size=14 (256 patches); per-image ρ is not directly comparable.

Metric. The primary metric is Spearman rank correlation (ρ) between pairwise cosine distances in patch embedding space and pairwise 3D Euclidean distances of patch centroids back-projected via camera intrinsics. We report *per-image* ρ (mean $\pm$ std over 654 val images, 19,110 pairs per image) and *cross-image* ρ (over all image-level pairs using mean-pooled embeddings and scene centroid 3D positions). All reported correlations are statistically significant.

A. Latent Geometry Analysis

Spatial-JEPA improves per-image ρ by **+123%** over I-JEPA (0.571 vs. 0.256) and cross-image ρ by **+316%** (0.104 vs. 0.025). DINOv2 achieves cross-image $\rho = 0.093$ via pretraining on 142M curated internet images, nearly matching Spatial-JEPA, but requires $178,000 \times$ more data. CLIP, despite 400M training pairs, scores below I-JEPA on per-image geometry (0.195 vs. 0.256), confirming that contrastive image-level training does not encourage patch-level metric structure.

Fig. 2 presents t-SNE visualisations coloured by metric depth. I-JEPA embeddings cluster by visual appearance; Spatial-JEPA embeddings form a smooth gradient from near (blue) to far (yellow).

B. Ablation: What Makes It Work?

Depth-aware masking alone provides no geometric improvement (0.255 vs. 0.256), confirming masking strategy is not the source of metric learning. Concatenation fusion improves cross-image ρ modestly (0.069 vs. 0.025) but barely affects per-image ρ (0.273). Symmetric cross-attention, the closest analogue to CMX-style RGB-D fusion [18] adapted to JEPA, improves only marginally over concatenation (per-image 0.282, cross-image 0.083). Asymmetric cross-attention yields the full improvement, isolating the role of the asymmetric query direction (depth attends to RGB) rather than the presence of attention itself:

Spatial-JEPA — Experiment 1: Latent Geometry Analysis

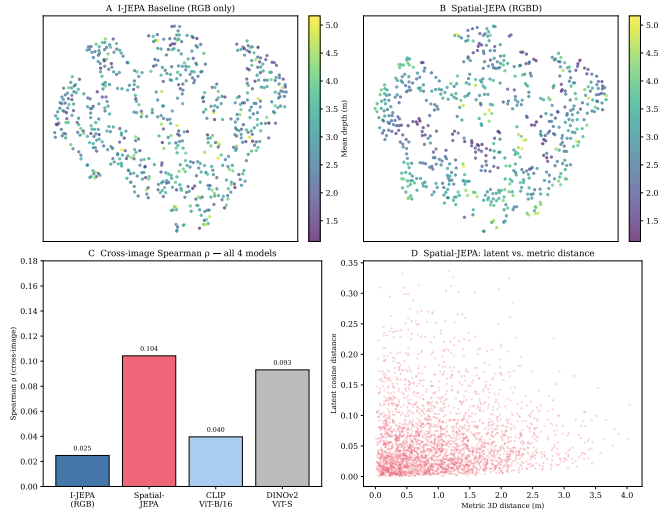


Fig. 2. t-SNE of patch embeddings (654 NYUv2 val images) coloured by metric depth (blue = near, yellow = far). *Top-left (A)*: I-JEPA, clusters by visual appearance, no depth gradient. *Top-right (B)*: Spatial-JEPA, smooth near-to-far continuum, patches at similar depths are nearby in representation space regardless of appearance. *Bottom-left (C)*: cross-image Spearman ρ across four models, showing Spatial-JEPA above all RGB-only baselines. *Bottom-right (D)*: per-pair scatter of latent cosine distance vs. metric 3D distance for Spatial-JEPA, with the positive trend that the per-image ρ summarises.

TABLE III

ABLATION STUDY, NYUv2 VAL. ALL FOUR RGB-D VARIANTS TRAIN UNDER THE IDENTICAL RECIPE; ONLY THE FUSION MECHANISM DIFFERS. PER-IMAGE ρ AND CROSS-IMAGE ρ

Variant	Per-image	Cross-image
I-JEPA Baseline (RGB, random mask)	0.256 ± 0.168	0.025
+ Depth-aware masking only (Abl. B)	0.255 ± 0.167	0.025
+ RGBD concat fusion (Abl. C)	0.273 ± 0.164	0.069
+ RGBD symmetric cross-attn (Abl. D)	0.282 ± 0.164	0.083
+ RGBD asym. cross-attn (Spatial-JEPA)	0.571 ± 0.166	0.104

ρ jumps from 0.282 to **0.571** per-image and from 0.083 to **0.104** cross-image.

C. Depth Estimation Probe

We train a lightweight probe head ($\sim 1.8k$ parameters: $\text{Conv}(384 \rightarrow 128 \rightarrow 1) + \text{bilinear upsample} + \text{sigmoid} \times 10$) on top of frozen encoders using SILog loss. **All four models receive RGB only at probe time**; Spatial-JEPA’s frozen encoder receives a zero depth tensor. This design is deliberate: providing real depth at probe time would give Spatial-JEPA a live signal unavailable to RGB-only baselines, conflating encoder quality with input advantage. The zero-depth condition isolates what the frozen representations *already contain*, geometry internalised during RGBD pretraining, now linearly readable without any depth input.

At 224×224 , Spatial-JEPA achieves **AbsRel = 0.046** vs. the best supervised alternative at 0.397, a $\sim 9 \times$ **improvement**. The three RGB/scratch models plateau at near-identical performance regardless of pretraining, confirming the bottleneck is the

TABLE IV

DEPTH ESTIMATION PROBE, NYUv2 VAL. BEST EPOCH OVER THE PROBE RUN. THE LOWER BLOCK RE-EVALUATES THE SAME FROZEN ENCODERS AT FULL 640×480 RESOLUTION BY BICUBIC-INTERPOLATING THE LEARNED POSITIONAL EMBEDDINGS TO A 30×40 TOKEN GRID.

Model	Encoder	Sup.?	AbsRel↓	RMSE↓	$\delta < 1.25 \uparrow$
<i>224×224 centre crops, 100-epoch probe</i>					
Spatial-JEPA (frozen)	RGBD SSL	No	0.046	0.280	0.974
I-JEPA (frozen)	RGB SSL	No	0.413	1.154	0.397
I-JEPA (fine-tuned)	RGB SSL	Yes	0.403	1.167	0.388
Supervised (scratch)	Random	Yes	0.397	1.217	0.359
<i>640×480 full resolution, 50-epoch probe</i>					
Spatial-JEPA (frozen)	RGBD SSL	No	0.037	0.222	0.983
I-JEPA (frozen)	RGB SSL	No	0.421	1.107	0.401

TABLE V

MECHANISTIC ANALYSIS, NYUv2 VAL. ALL MODELS USE RGB ONLY AT INFERENCE.

Model	Per-image ρ	Cross-image ρ
I-JEPA (100ep, RGB SSL)	0.256 ± 0.168	0.025
Spatial-JEPA (rgb_stream only)	0.246 ± 0.170	0.037
Null token, fixed vector	-0.117 ± 0.075	0.015
Surrogate, rgb→depth	0.114 ± 0.144	0.024

absence of geometric structure in the encoder, not model capacity.

Full-resolution evaluation. A reasonable concern about the 224×224 result is that it is not directly comparable to published NYUv2 SOTA evaluated at the standard 640×480 resolution (DPT-Hybrid [22] reports $\text{AbsRel} \approx 0.110$, training on 1.4M images with depth supervision). We re-evaluate both encoders at full 640×480 by bicubic-interpolating the learned positional embeddings to a 30×40 token grid (a standard ViT high-resolution transfer recipe [12]), train a fresh probe head, and observe that Spatial-JEPA *improves* to $\text{AbsRel} = 0.037$ while I-JEPA stays at 0.421. The relative gap grows from $9 \times$ to $11 \times$ at full resolution; the headline number is not an artefact of the lower resolution. We do not claim to match supervised SOTA: the comparison here is matched-regime (795 NYUv2 images, frozen encoder, 1.8k-parameter probe, no depth supervision), and isolates the contribution of the geometric prior in the encoder rather than pretraining scale.

D. Where Does the Geometry Live?

We run three RGB-only inference probes on the trained Spatial-JEPA encoder to ask whether the geometry has been baked into the RGB weights or whether it lives in the live depth signal. (A) *RGB stream isolation*: call `rgb_stream(rgb)` directly, bypassing the depth stream and cross-attention, a clean in-distribution test of the RGB weights alone. (B) *Fixed null token*: replace the depth stream output with a fixed zero vector and pass it through the frozen cross-attention (distinct from the *learnable* null token in Sec. III-D). (C) *RGB-to-depth surrogate*: a 2-layer transformer (3.7M params) is fine-tuned for 20 epochs to predict depth-stream tokens from RGB-stream tokens, and substituted for the depth stream at inference.

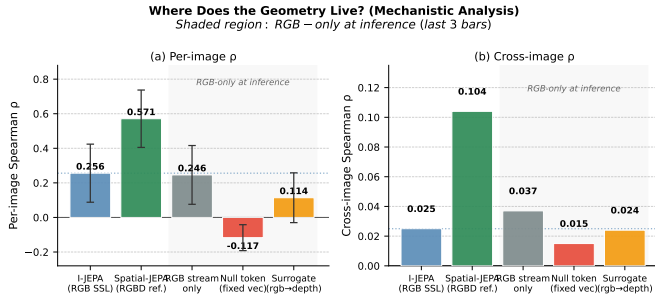


Fig. 3. Mechanistic analysis for four RGB-only inference conditions. *Left*: per-image ρ . The *rgb_stream* isolation result (green) sits directly on top of the I-JEPA baseline (blue), confirming RGB-D co-training leaves the RGB pathway geometrically unchanged. The null token (red) actively inverts geometry; the surrogate (orange) partially recovers it but remains below the baseline. *Right*: cross-image ρ for the same four conditions, in the same ordering; the picture is consistent.

All three probes converge on the same answer. RGB stream isolation gives $\rho = 0.246$, indistinguishable from the RGB-only I-JEPA baseline (0.256): RGB-D co-training did not substantively restructure the RGB weights. The null token actively inverts geometry (-0.117), and the RGB-to-depth surrogate cannot match the baseline (0.114). Geometry, in the embedding space, is depth-conditional: it is not recoverable from the RGB weights of the standard Spatial-JEPA encoder. Fig. 4 visualises the underlying mechanism: for a chosen depth-query patch, the cross-attention heatmap concentrates on RGB tokens at *similar metric depths* rather than visually similar tokens, which is how the depth signal organises the fused representation. The next subsection asks the corollary question, can we recover the prior *outside* the embedding space?

E. Depth-Free Deployment Paths

For platforms that do not have a calibrated depth sensor at inference time, we evaluate two routes that recover the geometric prior from RGB alone.

Route 1: monocular depth at inference. We replace the depth input to the trained Spatial-JEPA encoder with the output of a frozen pretrained monocular depth predictor (DepthAnything-v2-Small [23]). The Spatial-JEPA weights are unchanged; only the source of the depth tensor differs.

Route 2: cross-modal distillation. We train an RGB-only ViT student (depth=12, 384-dim, ~ 25 M parameters) to regress the trained Spatial-JEPA target encoder’s fused-token output via cosine-similarity loss for 200 epochs on NYUv2. A first attempt with a smaller student (depth=6, MSE loss, 100 epochs) collapsed (per-image $\rho = 0.130$, below the I-JEPA baseline); the deeper student with a directional loss recovers the prior.

Both routes recover most or all of the per-image geometric prior. The monocular-depth route reaches per-image $\rho = 0.578$, statistically indistinguishable from the GT-depth reference (0.571): Spatial-JEPA’s cross-attention is robust to the noise of a monocular predictor, and the combined system is deployable on RGB-only hardware. The distillation route recovers 79% of the per-image prior (0.454 vs. 0.571) and runs as a single ~ 25 M-parameter RGB-only ViT, with no inference-time dependency

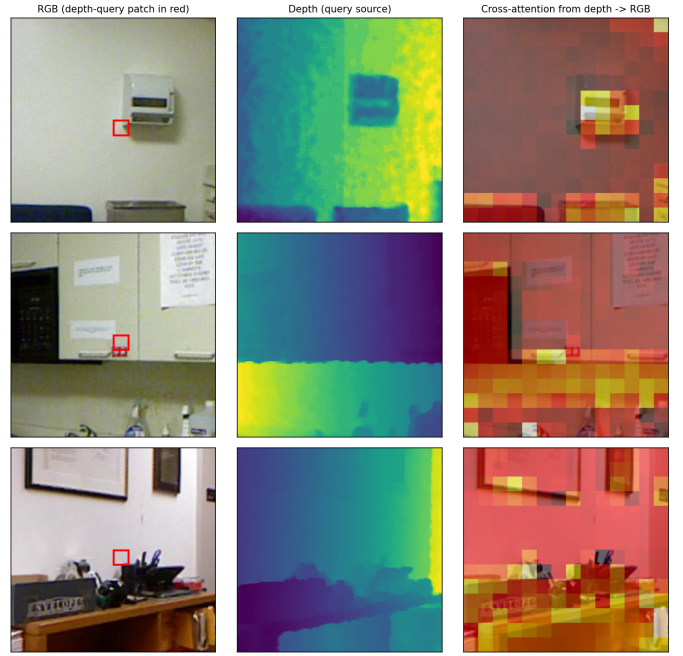


Fig. 4. Depth-to-RGB cross-attention. For a query depth patch (red box), the heatmap shows where in the RGB token grid the depth token attends. Bright regions correspond to RGB patches at metric depths similar to the query, not to patches that look visually similar.

TABLE VI
DEPTH-FREE DEPLOYMENT PATHS, NYUV2 VAL. THE ORIGINAL SPATIAL-JEPA INFERENCE PATH (RGB + GT DEPTH) IS SHOWN FOR REFERENCE. PER-IMAGE ρ AND CROSS-IMAGE ρ

Route	Inference input	Per-image	Cross-image
<i>Reference</i>	RGB + GT depth	0.571	0.104
Monocular	RGB + DepthAnything-v2	0.578	0.031
Distillation	RGB only (student)	0.454	0.012
Baseline	I-JEPA, RGB only	0.256	0.025

on a depth predictor. Cross-image ρ collapses in both routes; global scene structure is more sensitive to depth noise than local patch geometry, and the distilled student cannot fully reproduce the teacher’s scene-level representation. We report this honestly as a limitation: Spatial-JEPA’s local-geometric prior is robust to depth-free deployment, but its global-scene representation is not.

Combined with Sec. IV-D, the picture is consistent. Within the encoder weights, the geometric prior lives in the depth-conditional cross-attention output; outside the encoder, the prior can be relocated either to (a) the inputs (monocular depth prediction) or (b) a new student network trained against the original (distillation). The system as a whole has multiple deployment modes; the specific encoder we publish is one of them.

F. Depth Dropout Training

We train Spatial-JEPA from scratch for 100 epochs with $p = 0.5$ depth dropout (Sec. III-D), identical compute to the

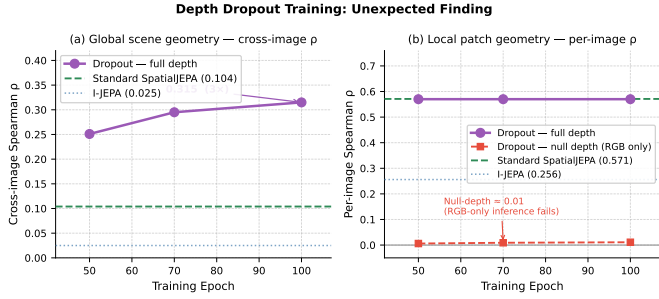


Fig. 5. Depth dropout training dynamics. Full-depth cross-image ρ (solid blue) climbs from 0.104 to 0.315 across 100 epochs. Full-depth per-image ρ (solid green) stays flat at 0.570, dropout does not degrade local geometry. Null-depth per-image ρ (dashed red) remains near zero throughout, confirming RGB-only inference is not recovered.

TABLE VII
DEPTH DROPOUT TRAINING, NYUV2 VAL. PROGRESS ACROSS EPOCHS.

Epoch / model	Full-depth per- ρ	Full-depth cross- ρ	Null-depth per- ρ
Standard Spatial-JEPA (100ep)	0.571	0.104	N/A
Dropout, ep. 50	0.570	0.251	0.006
Dropout, ep. 70	0.570	0.295	0.009
Dropout, ep. 100	0.570	0.315	0.011

standard model. **Full-depth mode** at epoch 100: per-image $\rho = 0.570$ (vs. 0.571 standard, negligible) but cross-image $\rho = 0.315$ vs. 0.104, a **3 $\times$ improvement** in global scene geometry at negligible cost to local geometry. **Null-depth mode**: per-image $\rho = 0.011$, near zero and flat across all epochs; RGB-only inference remains impossible.

Standard Spatial-JEPA can delegate all geometric signal to the depth stream, leaving the RGB pathway as pure appearance. When depth is randomly absent, the model must encode scene-level structure, room scale, depth distribution, into the RGB pathway to minimise the JEPA loss. When depth is present at inference, both pathways contribute: the result is richer global geometry than either pathway alone.

Depth dropout is not merely a robustness technique, it is a *better RGBD SSL training strategy* even when depth is always available at inference. We recommend $p = 0.5$ as the default configuration.

G. Zero-Shot Domain Transfer

We evaluate zero-shot on two unseen synthetic domains. **AI2-THOR** [14]: 600 RGBD frames from 60 scenes (kitchens, living rooms, bedrooms), rendered at 300×300 and resized to 224×224 . Intrinsic: $f_x = f_y = 194.0$, $c_x = c_y = 111.5$ (FoV = 60°). **HyperSim** [15]: 4,484 RGBD frames from 49 scenes. Depth stored as Euclidean ray length, converted to z -depth. Intrinsic: $f_x = f_y = 886.8$, $c_x = c_y = 112.0$.

On AI2-THOR, the per-image advantage narrows (0.287 vs. 0.283) because many simulated frames contain near-planar surfaces with low depth variance. Cross-image ρ reveals a qualitative difference: Spatial-JEPA stays positive (0.067) while I-JEPA goes negative (-0.017), indicating its global

TABLE VIII
ZERO-SHOT DOMAIN TRANSFER. ZS = ZERO-SHOT; FT = SIGREG FINE-TUNED ($\lambda=0.1$).

Model	Domain	Per-image ρ	Cross-image ρ
I-JEPA	NYUv2 (src)	0.256 ± 0.168	0.025
Spatial-JEPA	NYUv2 (src)	0.571 ± 0.166	0.104
I-JEPA	AI2-THOR (ZS)	0.283 ± 0.158	-0.017
Spatial-JEPA	AI2-THOR (ZS)	0.287 ± 0.183	0.067
Spatial-JEPA + SIGReg FT	AI2-THOR (FT)	0.342 ± 0.167	0.184
I-JEPA	HyperSim (ZS)	0.181 ± 0.219	0.160
Spatial-JEPA	HyperSim (ZS)	0.404 ± 0.230	0.195
Spatial-JEPA + SIGReg pretrain	HyperSim (ZS)	0.432 ± 0.217	0.399

TABLE IX
IN-DOMAIN SCALING, HYPERSIM (4,484 TRAINING FRAMES, 70 EPOCHS).

Model	Training data	Per-image ρ	Cross-image ρ
I-JEPA	NYUv2 (795 frames)	0.256 ± 0.168	0.025
Spatial-JEPA	NYUv2 (795 frames)	0.571 ± 0.166	0.104
I-JEPA	HyperSim (4,484 frames)	0.192 ± 0.216	0.244
Spatial-JEPA	HyperSim (4,484 frames)	0.439 ± 0.227	0.221

representations are anti-correlated with metric scene geometry in simulation.

SIGReg fine-tuning on AI2-THOR reaches cross-image $\rho = 0.184$, **surpassing the NYUv2 source level (0.104)**. On HyperSim, the SIGReg-pretrained model boosts zero-shot cross-image ρ from 0.195 to **0.399** (+104%), Gaussianising representations during pretraining makes all embedding dimensions uniformly informative.

H. Scaling to Larger RGBD Datasets

We train both models on HyperSim [15], 4,484 frames from 49 scenes in total (80/20 split: 39 training scenes, 10 val scenes), $6 \times$ NYUv2 in training data, for 70 epochs each.

Spatial-JEPA's per-image advantage is fully preserved at $6 \times$ scale: **+129%** on HyperSim (0.439 vs. 0.192) vs. +123% on NYUv2. The geometric prior from asymmetric cross-attention is not an artefact of small training sets. **Cross-Dataset Generalisation.** We evaluate HyperSim-trained checkpoints zero-shot on NYUv2.

Spatial-JEPA trained on HyperSim achieves $\rho = 0.611$ on NYUv2 zero-shot, *higher* than the model trained directly on NYUv2 (0.571). Cross-image ρ improves from 0.104 to **0.233** (+124%).

I-JEPA shows no cross-dataset benefit (0.243 vs. 0.256; 0.028 vs. 0.025). Critically, **both models train on identical HyperSim data**, the only difference is whether the model learns geometric structure or appearance statistics. Appearance statistics are domain-specific (HyperSim rendering $\neq$ NYUv2 photography); camera-relative 3D geometry is a physics-level invariant that holds regardless of rendering style, lighting, or camera rig.

TABLE X
CROSS-DATASET GENERALISATION, HYPERSIM-TRAINED $\rightarrow$ NYUV2 ZERO-SHOT.

Model	Training data	Eval domain	Per-image ρ	Cross-image ρ
I-JEPA	NYUv2 (795 frames)	NYUv2 (in-domain)	0.256 ± 0.168	0.025
Spatial-JEPA	NYUv2 (795 frames)	NYUv2 (in-domain)	0.571 ± 0.166	0.104
I-JEPA	HyperSim (4,484 frames)	NYUv2 (zero-shot)	0.243 ± 0.162	0.028
Spatial-JEPA	HyperSim (4,484 frames)	NYUv2 (zero-shot)	0.611 ± 0.154	0.233

TABLE XI

ZERO-SHOT TRANSFER OF NYUV2-PRETRAINED ENCODERS TO MEDICAL RGB-D DOMAINS. SIMCOL3D: PHOTOREALISTIC SYNTHETIC COLONOSCOPY (475 $\times$ 475, 37K FRAMES, DEPTH RANGE 0–20 CM). C3VD: REAL PHANTOM-TISSUE COLONOSCOPY WITH SPECULAR HIGHLIGHTS AND LENS DISTORTION (1350 $\times$ 1080, 12 SEQUENCES, DEPTH RANGE 0–10 CM).

Domain	Model	Per-image ρ	Cross-image ρ
SimCol3D (synth.)	I-JEPA	0.618 ± 0.129	0.848
	Spatial-JEPA	0.666 ± 0.090	0.821
C3VD (real)	I-JEPA	0.088 ± 0.068	0.784
	Spatial-JEPA	0.223 ± 0.093	0.767

Practical implication. Spatial-JEPA can be pretrained on large-scale synthetic RGBD data (cheap and unlimited to generate) and deployed on real scenes with geometric structure that meets or exceeds models trained on expensive real-world RGBD data. The bottleneck is synthetic scene diversity, not real data collection.

I. Downstream Transfer to Medical RGB-D

To stress-test the geometric prior on a deployment-relevant domain, we zero-shot transfer the NYUv2-pretrained encoders to two endoscopic RGB-D datasets (Table XI). On C3VD, real phantom tissue with specular highlights, motion blur, and lens distortion that fall outside any indoor training distribution, Spatial-JEPA’s per-image ρ is $2.5\times$ the I-JEPA baseline (0.223 vs. 0.088) despite never having seen colonoscopy imagery. On SimCol3D, both encoders cluster in the high- ρ regime (> 0.6) because the synthetic colon’s tube geometry produces strong global depth gradients that even appearance features partially track; the per-image gap (Spatial-JEPA 0.666 vs. I-JEPA 0.618) confirms the geometric prior still wins on local structure. This is direct evidence that the contribution generalises to a downstream RGB-D system that bears no resemblance to the pretraining distribution.

Reverse transfer: medical pretraining $\rightarrow$ indoor scenes.

We also evaluate the reverse direction. A Spatial-JEPA encoder pretrained on SimCol3D (37k synthetic colonoscopy frames, no labels) and zero-shot transferred to NYUv2 reaches per-image $\rho = 0.585$, exceeding the NYUv2-trained model at 0.571. The matched I-JEPA baseline SimCol3D-pretrained on NYUv2 lands at 0.266, essentially the in-domain NYUv2 baseline of 0.256, showing no cross-domain benefit. Combined with the HyperSim-to-NYUv2 result of Sec. IV-H, we have two independent synthetic-source datasets that transfer to real indoor scenes at or above in-domain training performance, while RGB-only appearance encoders show no such transfer.

V. DISCUSSION

I-JEPA vs. Spatial-JEPA. I-JEPA encodes: “*What does this patch look like?*”, texture, colour, semantic category. Spatial-JEPA encodes: “*Where is this patch in 3D space?*”, metric position relative to the camera. The depth probe result (AbsRel 0.046 vs. 0.397) quantifies the gap: for tasks requiring metric spatial reasoning, the geometric prior provides substantially stronger signal than appearance-only SSL, even when RGB models are fully supervised with labelled depth.

Depth at inference: a design choice with multiple deployment paths. Spatial-JEPA is targeted at platforms that already carry calibrated depth sensors: ToF and structured-light cameras on mobile robots, stereo rigs on manipulators, depth-from-defocus on endoscopes, LiDAR-camera fusion stacks, and RGB-D SLAM front-ends. For these systems, depth is not an extra input but a primary signal that is already paid for; the question is what representation makes use of it. For RGB-only hardware, Sec. IV-E demonstrates two practical routes that recover the geometric prior: chaining the trained encoder with a frozen monocular depth predictor (zero training, full recovery on per-image ρ) and cross-modal distillation into an RGB-only student (one training run, 79% recovery, no inference-time dependency on a depth predictor). Our mechanistic experiments (Sec. IV-D) clarify why a naive RGB-only path fails: in the standard Spatial-JEPA encoder the geometric prior is depth-conditional in the embedding space, so it must be relocated either to the inputs (monocular prediction) or to a new network (distillation) for depth-free deployment.

Depth dropout as a general principle. The $3\times$ cross-image improvement suggests a broader principle: *modality dropout forces encoding of shared information into all available pathways*. Future work could apply this to audio-visual, RGB-thermal, and RGB-polarisation settings.

Limitations. (1) Real RGBD scale: 795 real training images is small; experiments on ScanNet [16] and ARKitScenes [19] would better characterise limits. The cross-dataset result (Table X) partially addresses this, synthetic HyperSim pretraining transfers *better* than real NYUv2 pretraining, suggesting synthetic data is a viable substitute. (2) Depth-free deployment routes are partial: both the monocular-depth and distillation routes recover most of the per-image prior but cross-image ρ collapses (Table VI). Global scene structure remains best preserved with a real depth sensor at inference. (3) Custom metric: Spearman ρ is not a standard benchmark; the depth probe (AbsRel=0.046 vs. 0.397) and the zero-shot medical RGB-D transfer (Sec. IV-I) provide standard-task downstream

evidence, but benchmarking on additional robotic perception tasks (object 6-DoF pose, voxel occupancy prediction) is the direct next step.

VI. CONCLUSION

We introduced Spatial-JEPA, a JEPA variant that learns metric 3D geometry from scene-level RGBD data via asymmetric cross-attention fusion. Key findings:

- Per-image $\rho = 0.571$ on NYUv2 vs. 0.256 for I-JEPA, with Ablation D (symmetric cross-attention) at 0.282 confirming the asymmetric query direction is the decisive design choice.
- Frozen depth probe AbsRel=0.046 at 224×224 and **0.037** at full 640×480 resolution ($11\times$ a matched-regime RGB-only baseline).
- Mechanistic experiments locate the geometric prior in the depth-conditional cross-attention output, not in the RGB weights.
- Two depth-free deployment routes recover the prior on RGB-only hardware: chained monocular depth at inference matches the GT-depth result ($\rho = 0.578$); cross-modal distillation into an RGB-only student recovers 79% of the per-image prior ($\rho = 0.454$).
- Cross-domain pretraining transfers cleanly: HyperSim-trained Spatial-JEPA achieves $\rho = 0.611$ on NYUv2 zero-shot (vs. 0.571 for the NYUv2-trained model), and SimCol3D-trained achieves $\rho = 0.585$ (also above NYUv2-trained), while RGB-only baselines show no cross-domain benefit.

Future work will evaluate on ScanNet and ARKitScenes, scale to ViT-Base, and explore patches defined by image segmentation rather than fixed grid cells as the spatial primitive for the encoder.

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